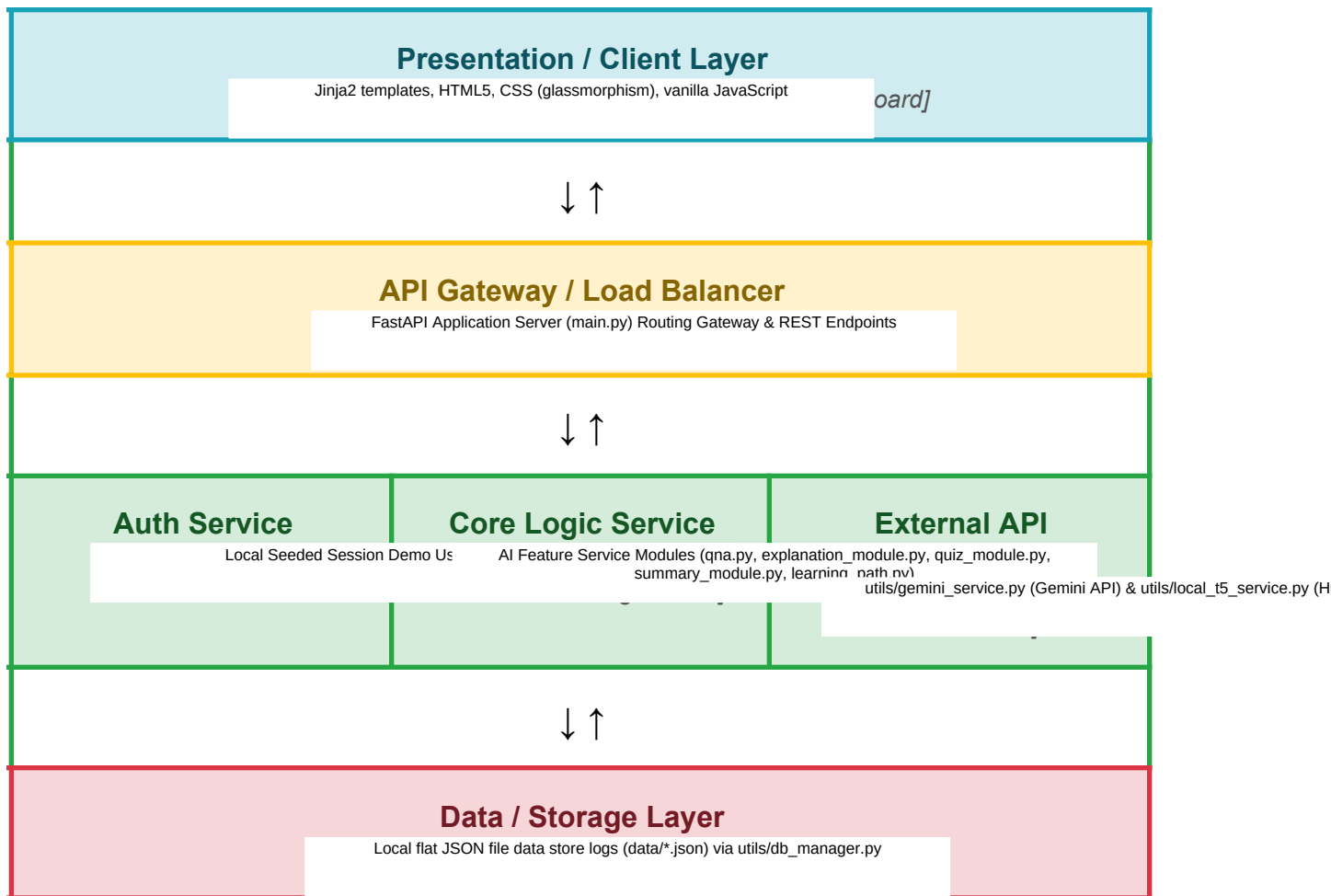


Solution Architecture

Date	28 June 2026
Team ID	6a3a555ae5895b2c9759029d
Project Name	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	Renders the home, chat, explain, quiz, summary, recommend, about and contact pages; handles client-side interactivity.	Jinja2 templates, HTML5, CSS, vanilla JS
<i>[API Gateway]</i>	FastAPI app (main.py) exposing router-based endpoints (/api/ask, /api/explain, etc.) and direct RESTful endpoints.	FastAPI, Uvicorn, CORS middleware
<i>[Microservices]</i>	Feature-specific modules (qna.py, explanation_module.py, etc.) that build prompts, call Gemini/T5, and handle fallback.	Python, Pydantic schemas (models/schemas.py)
<i>[Database]</i>	utils/gemini_service.py dynamically resolves best model; utils/local_t5_service.py lazily loads T5 in background thread.	google-generativeai SDK, Hugging Face transformers, PyTorch